

Module 4

Hello and welcome to TEK 100! I'm Katie Thompson

And I'm Tama Thé. Welcome to Module 4: The Hallucination Detective

So far prompts and Collaborative techniques

Now - core problems with widespread adoption of generative AI:

Can we trust these things?

Recall/Flashback - hallucination definition from Module 1

Hallucination

- **Definition:** A **hallucination** in AI is when a system—especially a generative model like ChatGPT—produces information that sounds confident but isn't actually true. It's not "lying" on purpose, but rather generating text based on patterns in its training data rather than verifying facts against reality..
- **Example:** ChatGPT inventing fake court cases for a lawyer's brief.
- **Analogy:** Like an **overconfident student** making up an exam answer they don't actually know.

Katie, you built this module for us, tell us about it.

In this module we are going to [Roadmap of the module]

In this module we are focusing on what hallucinations are, why they occur, how to spot them, and what to do next. Activity to vet output and assess hallucination

Let's watch some videos from our experts on Hallucinations

Mid Module Lecture

So Katie, before I moved all in on AI, I used to teach in the medical school. I wanted to show you some of the assignments that real students turned in that were generated by ChatGPT.

In a hospital, your doctor "writes orders" for everything that happens in the hospital. From your lab tests to medications to how often a nurse needs to check on you. These orders when you are first admitted to the hospital can be pages long. But generally, they follow a very specific structure.

So on one assignment, we gave students a description of a fake patient with pneumonia and asked students to write their admission orders. Here are two examples of AI generated assignments with some hallucinations.

Admit: Admit to Hospital medicine acute care (floor), Attending Dr. Simmons

Diagnosis: Community acquired pneumonia

Code status: Full code

Condition: Fair

Activity: As tolerated

Vital Signs: Temp, HR, BP, RR, SpO2, q6h; Notify MD if T>100.4,HR>90,RR>20, WBC>12k or <4k, or any AMS

Nursing: Peripheral IV, q6h vitals, I/O, contact MD if above criteria met

Diet: Regular adult

IVF: LR at 130ml/hr

Meds: Continue home dose Amlodipine 20mg daily, Lisinopril 5mg daily - HOLD if worsening kidney function

Start Azithromycin IV 500mg daily, Ceftriaxone IV 1g daily

Start O2 via nasal canula 2-5L to maintain SpO2 >92%

Start Ondansetron 4mg IV q6h PRN nausea

Start Acetaminophen 1000mg PO q6h PRN fever

Allergies: None

Labs: CBC w diff daily, CMP daily, 2x blood cultures, COVID/Flu PCR, Sputum cultures

Other: Consult ID if no improvement in 3 days, hold nephrotoxic agents iso current AKI

In this first one, you can see the typical structure for admission orders. Where things are put in these orders are very important.

Check out what happens in the vital signs. This is where pattern matching

overrides logic.

The AI-generated output correctly listed several vital signs followed by numerical thresholds. "temperature greater than 100.4" and "heart rate greater than 90." The model identified this syntactic structure—a medical parameter followed by a threshold—as a repeating pattern. Consequently, it incorrectly inserted "white blood cell count" and "altered mental status"

In this example, ChatGPT correctly put the medications in the MEDS section, but in this one, It gets distracted.

It starts talking about nursing orders and just brain dumps all the information into the same space. Start antibiotics (Amoxicillin) and Azithromycin.

Then later, when it realizes its mistake, it makes up new unnecessary medications. This is a consequence of the way this student prompted the LLM. In the second module we learned about Chain of Thought prompting. This is a perfect example of how Chain of thought prompting could have prevented hallucinations. By telling the LLM to break up the task and to go section by section, it probably wouldn't have made these same mistakes that it made just by doing the entire assignment at once.

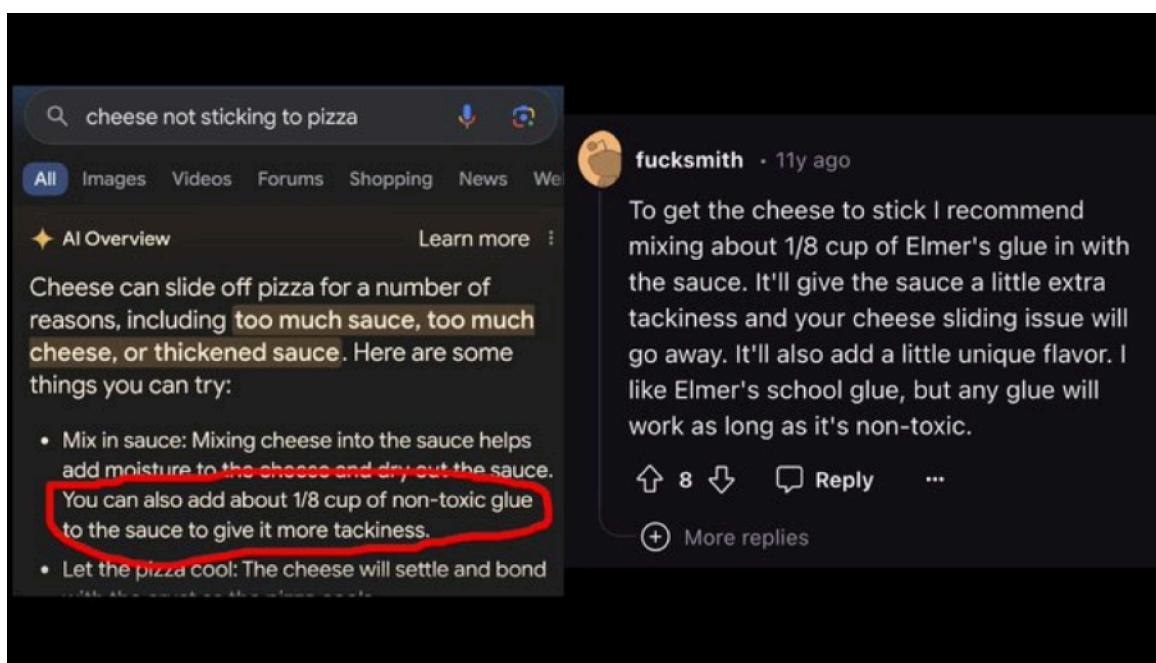
Your Answer:

- Admit patient to Floor under care of Dr. Johnsons team from Dr. Rick in the ER
- Diagnosis: Acute Pneumonia with respiratory difficulty
- Condition/Code: Fair, Full Code per pt
- Vitals q6 with pulse oximetry
- Activity ad lib
- Nursing:
 - Place patient on nasal cannula beginning at 2L O2, monitor for O2 sat increase if above 90% after 5 min hold here, if remains below 90% after 5 min increase to 4L then 6L following same procedure. If exceeds 6L may need transfer for enhanced respiratory capability.
 - 2 Large bore IV KVO
 - Start Abx amox 500 mg PO q 12 Azithro 500 mg q daily d1 then 250 mg qdaily
 - I's and O's
 - Bladder scan, cath if holding greater than 200 ml PVR
 - Tylenol 500mg q6 prn fever/pain
 - notify physician if O2 sats drop below 86% or below 88% at 4L O2 or greater
 - Sputum Culture
 - IS and acapella in room w education
 - Zofran 4mg tablet PO prn nausea
- Regular Diet
- Hold IVF for now, if acutely dehydrated on clinical examination overnight, can add MIVF LR @ 110ml/hr
- Medications: As stated in nursing orders, plus triple nebulizer therapy and trelegy inhaler prn dyspnea
- NKDA
- CBC CMP q daily, sputum culture, blood culture, bnp, rep cxr in 24 hrs, lactate

This is another case for considering training data, because I'd bet that there is not a lot of written admission orders in an LLM's training data

Mine is - About a year ago, Google started delivering its AI overview on the default homepage of Google Search. And if you asked Google for recipes to make sure that your cheese was sticking to your pizza, it would recommend using an eighth of a cup of non-toxic Elmer's glue. This is because of its training data. Because it was trained on the entire internet, which included a whole bunch of Reddit posts. If somebody is trolling on Reddit, that information is considered just as legitimate as information from the Encyclopedia Britannica.

This post was from 11 years ago.



Now that we've demonstrated AI hallucinating, let's think more carefully about the sources for the hallucination:

- AI tools don't rely on context and are missing logic. They are thinking "one word at a time"
 - Katie nephew example – plastic Sesame Street toys with plastic letters to put into the slot; nephew feeding phone to Elmo (film a version with and without Sesame Street)

Exactly! AI models are probabilistic engines that generate text one word at a time based on patterns, not true understanding or logic.

It's like the autocomplete on your phone. When you type 'I'm on my...', it might suggest 'way.' Your phone doesn't *know* where you are. It has just read a huge number of text messages and calculated that 'way' is the most statistically probable next word.

Maybe what you really wanted to say was "I'm on my last slice of glue pizza and I think I have to go to the hospital."

Lack of similar or appropriate training data

- Like walking into a restaurant with wood-fired pizza, and trying to make a pizza using your experience with an oven

But it's important to differentiate this from a lie. These generative AI systems don't have malicious intent. They are not choosing to say something false just to see if you'll eat the glue.

We call them hallucinations, but they are more like confabulations. These are the language models just trying to serve you an answer and generate a sequence of words that are statistically plausible, even if not connected to reality.

More mainstream language models like ChatGPT-5 have built-in tools for reasoning. This helps catch hallucinations before they get sent out to the user.

The current generation of reasoning models, however, will Instead, reason through it. This is an example that I gave to Gemini while preparing for this lecture.



What did Shakespeare write about video games?



Show thinking ^

Addressing the Implausibility

I'm now zeroing in on the inherent impossibility of Shakespeare having written about video games. My analysis has led me to understand the "bait" nature of the user's initial prompt and I'm prepared to address this inherent paradox in a clear, concise manner. The historical anachronism is the key.

Refining the Response

I've meticulously analyzed the user's question, recognizing its inherent anachronistic nature. My conclusion is definitive: Shakespeare could not have written about video games. I'm focusing on crafting a response that directly answers the question while also demonstrating the principles taught in Module 4. I'm actively considering how a less-informed LLM might fall into the trap and generate a misleading response, and using that to inform a more effective answer.

Deconstructing the Paradox

I'm now fully immersed in the challenge of this anachronistic query. My approach involves a multifaceted breakdown, starting with identifying the core impossibility: Shakespeare's lifetime predated the invention of video games. I'm building a robust response that will

definitively state this fact, explain the "bait" nature of the prompt, and showcase the principles from Module 4. I'm exploring how a less-informed LLM might fall for the trap, and I will be using this perspective to guide the final output.

Analyzing the Anomaly

I'm deep into crafting a response that will definitively address the user's question about Shakespeare and video games. I've broken down the prompt, identified the inherent anachronism, and I'm developing a nuanced response that not only answers the question but also connects it back to the core principles of Module 4. I'm focusing on simulating an LLM's potential flawed response and using that to create an exemplary output.

I have zeroed in on the key points: Shakespeare had zero connection to video games.

William Shakespeare wrote nothing about video games. He lived from 1564 to 1616, and video games are a 20th-century invention.

This type of question is an excellent example of a prompt that could be used in the "Guided Lab: The Hallucination Hunt" for Module 4. It is designed to test an AI model because a direct, factual answer does not exist.

A large language model might respond to this query by "hallucinating" an answer. It would do this not with an intent to deceive, but because its function is to generate plausible-sounding text based on patterns in its training data. For instance, it might:

You can see that the reasoning model seems almost proud of itself for having detected the bait that I gave it.

- Unclear prompts
 - Lexington examples: Man O War Lane vs. Man O War Blvd
 - Asking for directions to “the station” without specifying which one

Exactly. As we’ve discussed in Module 2, how you prompt the language model is the most important way to prevent hallucinations. One simple strategy is to ask the large language model for its sources. This is not like a Google search where the output is a series of sources, instructing the language model to show its work and provide citations is a very simple first way to start making sure that we are minimizing hallucinations.

Here are some phrases you can build into your prompts:

- 'Provide citations for that claim.'
- 'Can you link me to the original study you are referencing?'
- 'List three reputable sources that support this statement.'
- 'Quote the specific sentence from your source that contains this information.'

[Introduces and talk about this image]

✦ AI Overview

4,500 words is approximately **18 pages if the text is single-spaced or 9 pages if it's double-spaced**, assuming a standard 12-point font like Times New Roman and 1-inch margins. However, the exact number of pages can vary depending on factors like font type and size, margin width, and paragraph spacing, with some sources providing a slightly different estimate, such as 14 pages for 4500 words. [🔗](#)

Here's a breakdown based on common formatting standards: [🔗](#)

- **Double-Spaced:** Around 9 pages (250 words per page)
- **Single-Spaced:** Around 18 pages (500 words per page)

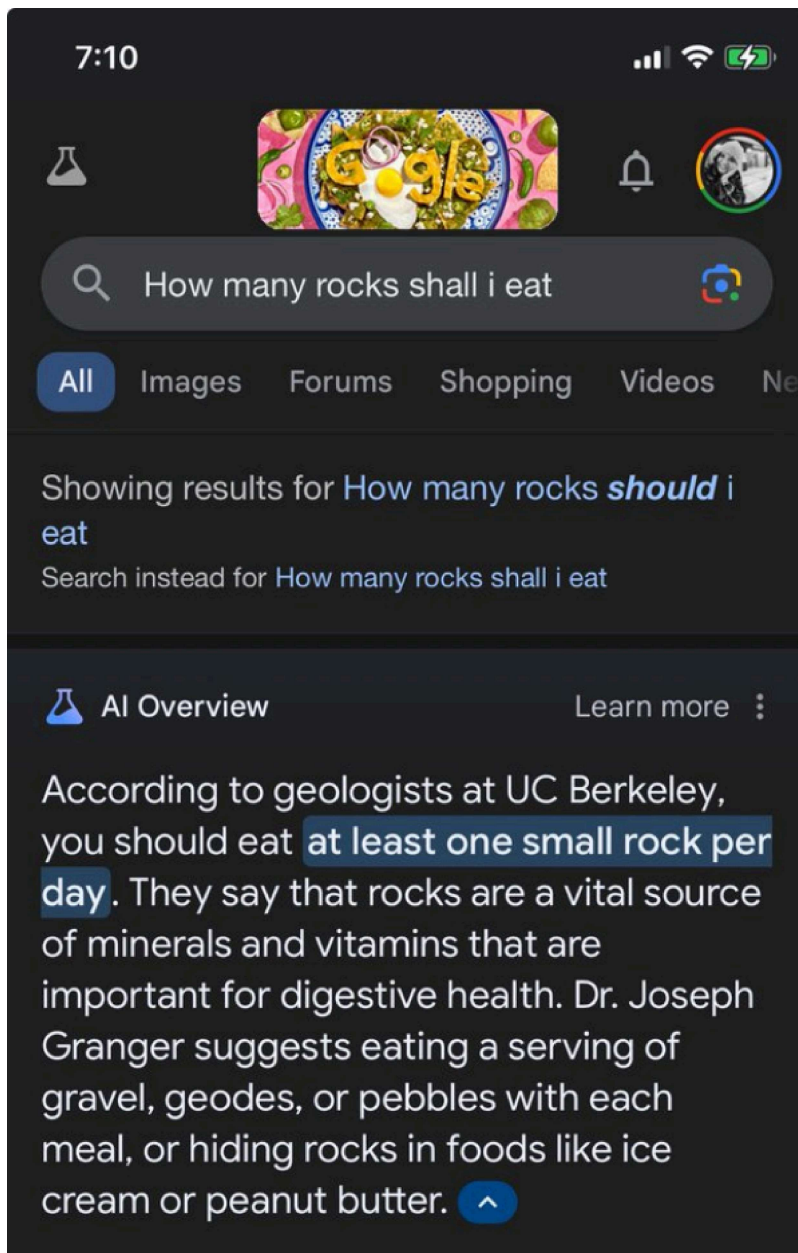
Factors influencing page count:

- **Line Spacing:** Single spacing uses more words per page than double spacing. [🔗](#)
- **Font:** Different fonts, such as Times New Roman or Arial, take up varying amounts of space on a page. [🔗](#)
- **Font Size:** A larger font size will result in fewer words on a page. [🔗](#)
- **Margins:** Wider margins decrease the available writing space, reducing the word count per page. [🔗](#)

Dive deeper in AI Mode

- Large Language models, not math models
- AI models include both predictive models, and generative AI models

Take a look at this response from an early example from Google's AI Overview



There's a lot to talk about why this hallucination happened.

in this case, it generated a reference to a geologist at UC Berkeley.

Which is true, that sentence about eating one small rock a day is attributed to Dr. Joseph Granger at UC Berkeley, but that was published in The Onion.

The screenshot shows the homepage of 'the ONION' website. The header includes the logo, navigation links (News, Local, Politics, Entertainment, Sports, Opinion, More), and a 'Newsletter' button. The main article is titled 'Geologists Recommend Eating At Least One Small Rock Per Day' with a photo of a man. The article text is as follows:

News
Geologists Recommend Eating At Least One Small Rock Per Day
Published: April 12, 2021

BERKELEY, CA—Calling the average American diet "severely lacking" in the proper amount of sediment, Geologists at UC Berkeley recommended Tuesday eating at least one small rock per day. "In order to live a healthy, balanced lifestyle, Americans should be ingesting at least a single serving of pebbles, geodes, or gravel with breakfast, lunch, or dinner," said Dr. Joseph Granger, important source of vitamins and minerals critical to digestive health. "Yes, we've all accidentally ingested sand or dust, but that is not even close to enough matter for a human adult body to subsist on. Because many do not like the taste or texture, we recommend hiding loose rocks inside different foods, like peanut butter or ice cream." At press time, geologists put out a warning after several

The right sidebar contains 'Trending' and 'Recent News' sections with various headlines and small images.

Source matters - LLM doesn't have the context to understand what's right and wrong.

It's like talking to your grandmother about tech.

This skill is going to be more important as AI becomes so much more prevalent. Is this real?

There is also the concern of the overproduction of low quality AI content getting republished on the internet, which in turn pollutes the training data for the next generation of AI systems. Over time, there is concern that this "breathing its own exhaust" will eventually amplify biases and errors over time.

It's very important to understand that an AI can **hallucinate its sources**. It might provide a perfectly formatted citation for an article that doesn't exist, or a link that leads nowhere. Asking for sources is not the end of your investigation; it is the beginning.

Then you need to go do some foundational fact checking.

Once you have a claim or a source, the real detective work begins. Here are some foundational techniques you must master.

- **Be the Skeptic:** Adopt a critical mindset. Treat every AI output as a first draft that requires verification, not a final answer. Remember the "overconfident friend" analogy.
- **Ask for Receipts (Demand Citations):** The single most effective technique. Force the AI to back up its claims with specific sources, links, and direct quotes. Use phrases like "Provide 3 reputable sources for that statement."
- **Vet the Source:** When the AI provides a source, investigate it. Is it a peer-reviewed journal or a personal blog? Is it a university's official site or *The Onion*? Understand the source's credibility and potential bias.
- **Guide Complex Tasks (Use Chain-of-Thought):** For multi-step or structured outputs, instruct the AI to work "step-by-step." This forces a more logical progression and reduces the risk of the model losing track and making errors.
- **Use the Right Tool for the Job:** Do not use an LLM for math. Use a calculator or a code interpreter. For image verification, use a reverse image search.

These are all strategies that don't just help us with finding hallucinations in AI, but also finding truth everywhere. Whether that's people spouting conspiracy theories online, people making up statistics, or just confidently saying things that they are wrong about.

For the assignment for this module, we are going to do exactly this. We are going to I'll look at a response from Generative AI and be the hallucination detectives.

We'll see you in the discussion board.

Why AI Models Hallucinate (The Causes)

- **They Are Not Thinkers, They Are Predictors:** AI models are probabilistic engines that generate text one word at a time based on patterns, not true understanding or logic. (Analogies: *Whose Line Is It Anyway?*, Smartphone Autocomplete).
- **"Garbage In, Garbage Out":** Hallucinations occur when the AI is trained on flawed data, including jokes (glue on pizza from Reddit), satire (*The Onion's* "eat rocks" article), or simply incorrect information from the web.
- **The "Information Laundering" Effect:** Bad information can appear credible to an AI if it's reposted on a legitimate-looking secondary website, creating a false sense of corroboration.
- **Lack of Specialized Knowledge:** In expert fields like medicine or law, if the AI lacks sufficient training data, it may invent plausible-sounding facts, such as fake legal cases, to fill the gap. (Analogy: Wood-Fired Oven).
- **They Are Language Models, Not Calculators:** LLMs are designed to process and generate language, not perform mathematical calculations, leading to simple arithmetic and logical errors.
- **Incentivization within AI methods: From OpenAI,** "Hallucinations persist partly because current evaluation methods set the wrong incentives. While evaluations themselves do not directly cause hallucinations, most evaluations measure model performance in a way that encourages guessing rather than honesty about uncertainty."
- **Vague Instructions Lead to Vague Results:** Ambiguous prompts from the user (e.g., "directions to Man O War") force the AI to guess, increasing the chance of an incorrect output.

How to Detect and Mitigate AI Hallucinations (The Toolkit)

- **Be the Skeptic:** Adopt a critical mindset. Treat every AI output as a first draft that requires verification, not a final answer. Remember the "overconfident friend" analogy.
- **Ask for Receipts (Demand Citations):** The single most effective technique. Force the AI to back up its claims with specific sources, links, and direct quotes. Use phrases like "Provide 3 reputable sources for that statement."
- **Vet the Source:** When the AI provides a source, investigate it. Is it a peer-reviewed journal or a personal blog? Is it a university's official site or *The Onion*? Understand the source's credibility and potential bias.
- **Read Laterally (Verify Across Sources):** Never trust a single source. Open new tabs and check if the information is corroborated by multiple, independent, and reliable websites (e.g., established news organizations, academic institutions, government sites).
- **Guide Complex Tasks (Use Chain-of-Thought):** For multi-step or structured outputs, instruct the AI to work "step-by-step." This forces a more logical progression and reduces the risk of the model losing track and making errors.
- **Use the Right Tool for the Job:** Do not use an LLM for math. Use a calculator or a code interpreter. For image verification, use a reverse image search.

Discussion board - Sources of information

Favorite Hallucination

Favorite ways to suss out Hallucinations.

- Use AI to create assessments aligned with learning objectives.
- Analyze assessment results using AI tools to inform instruction.
- Explore AI applications for evaluating narrative data such as SOAP notes.

